

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the construction and working and mathematical formula for discharge for venturimeter.

Q.24 Derive Hagen-Poiseuille's equation for laminar flow in a circular pipe. What are its limitations?

Q.25 Write short note on any two:-

- a) gate valve
- b) steady and unsteady
- c) priming

No. of Printed Pages : 4

220527

Roll No.

2nd Sem / Chemical, Chem P&P

Subject : Fluid Flow

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Bernoulli's theorem is based on the principle of conservation of:

- a) Mass
- b) Momentum
- c) Energy
- d) Temperature

Q.2 Laminar flow typically occurs at Reynolds number:

- a) Less than 2000
- b) Greater than 4000
- c) Between 2000 and 4000
- d) Equal to 5000

Q.3 Pitot tube is used to measure:

- a) Static pressure
- b) Velocity of flow
- c) Flow temperature
- d) Volume of fluid

Q.4 Cavitation in pumps is caused by:

- a) Low velocity
- b) High temperature
- c) Low pressure at suction
- d) Excessive priming

Q.5 Which valve is best suited for throttling purpose?

- a) Gate valve b) Butterfly valve
- c) Globe valve d) Ball valve

Q.6 Which law follow by Newtonian fluid?

- a) Bernoulli's
- b) Newton's law of viscosity
- c) Pascal's
- d) Hooke's

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A fluid is said to be incompressible if its _____ remain constant.

Q.8 Venturimeter is used to measure the _____ of fluid.

Q.9 Bernoulli's equation is applicable for _____ flow.

Q.10 Define Pump?

Q.11 Write the formula of Reynold number.

Q.12 Full form of BWG

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between the Uniform and Non-uniform flow with suitable examples.

Q.14 State Pascal's law with derivation

Q.15 Explain the working of a reciprocating pump with a neat diagram.

Q.16 Derive the expression for friction factor equation.

Q.17 Explain the construction and working of a Rotameter.

Q.18 Discuss the colour coding of industrial pipping used for transportation of various fluids.

Q.19 Differentiate between laminar, turbulent flow and transition flow

Q.20 Explain frictional losses in pipes.

Q.21 List various types of pipe fittings used in industry.

Q.22 List and explain any two type of manometer.